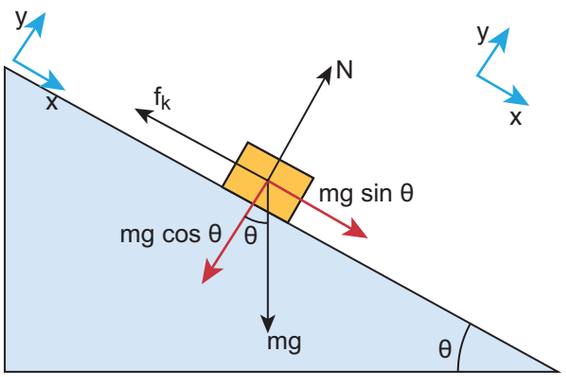
EXAMPLE 3.20

A block of mass m slides down the plane inclined at an angle 60° with an acceleration $\frac{g}{2}$. Find the coefficient of kinetic friction?

Solution

Kinetic friction comes to play as the block is moving on the surface.

The forces acting on the mass are the normal force perpendicular to surface, downward gravitational force and kinetic friction f_k along the surface.



Along the x-direction

$$mg\sin\theta - f_k = \text{ma}$$

But a = g/2

$$mg\sin 60° - f_k = \text{mg}/2$$

$$\frac{\sqrt{3}}{2}\text{mg} - f_k = \text{mg}/2$$

$$f_k = mg\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)$$

$$f_K = \left(\frac{\sqrt{3}-1}{2}\right)\text{mg}$$

There is no motion along the y-direction as normal force is exactly balanced by the mg $\cos\theta$.

$$\text{mg}\cos\theta = \text{N} = \text{mg}/2$$

$$f_K = \mu_K \text{N} = \mu_K \text{ mg}/2$$

$$\mu_K = \frac{\left(\frac{\sqrt{3}-1}{2}\right)mg}{\frac{mg}{2}}$$

$$\mu_K = \sqrt{3}-1$$

3.6.7 Application of Angle of Repose

1. Antlions make sand traps in such a way that when an insect enters the edge of the trap, it starts to slide towards the bottom where the antilon hide itself. The angle of inclination of sand trap is made to be equal to angle of repose. It is shown in the Figure 3.30.



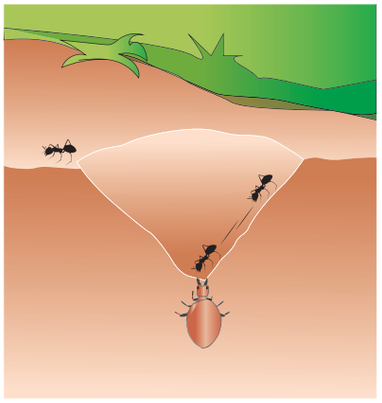


Figure 3.30 Sand trap of antlions

2. Children are fond of playing on sliding board (Figure 3.31). Sliding will be easier

UNIT-3(XI-Physics_Vol-1).indd 143 28-01-2020 13:48:38